

Rinky Sharma

Decodelabs Data Analytics Internship

Project: 1

Data Cleaning & Preparation**

Goal:

Clean a raw dataset by handling missing values, duplicates, and incorrect data.

Key Requirements:

Identify missing or null values

Remove duplicates

Correct data formats (dates, numbers, text)

Key Skills:

Data cleaning, Excel or Python basics, data preparation

```
In [11]: import pandas as pd

# Load your Excel dataset
df = pd.read_excel(r"C:\Users\Rinky Sharma\Downloads\Dataset for Data Analytics.xlsx")

# 1. Identify missing or null values
print("Missing values per column:")
print(df.isnull().sum())

# Example: Fill missing values (forward fill)
df = df.ffill()
# Or drop rows with missing values
# df = df.dropna()
```

Missing values per column:

OrderID	0
Date	0
CustomerID	0
Product	0
Quantity	0
UnitPrice	0
ShippingAddress	0
PaymentMethod	0
OrderStatus	0
TrackingNumber	0
ItemsInCart	0
CouponCode	309
ReferralSource	0
TotalPrice	0

dtype: int64

```
In [7]: # Remove duplicates
df = df.drop_duplicates()
print(df)
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	\
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	
...	...	...	...	...	...	...	
1195	ORD201195	2024-06-20	C21126	Desk	1	107.04	
1196	ORD201196	2024-03-04	C20095	Monitor	2	662.53	
1197	ORD201197	2023-07-13	C79674	Tablet	2	436.84	
1198	ORD201198	2024-08-22	C64753	Chair	4	262.52	
1199	ORD201199	2023-06-11	C57502	Tablet	4	560.58	

	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	\
0	928 Main St	Debit Card	Shipped	TRK37947903	7	
1	823 Main St	Online	Shipped	TRK91186779	3	
2	512 Main St	Credit Card	Cancelled	TRK42903982	8	
3	275 Main St	Debit Card	Returned	TRK62788070	5	
4	668 Main St	Online	Delivered	TRK29241424	8	
...	...	...	...	...	...	
1195	392 Main St	Credit Card	Cancelled	TRK38009181	6	
1196	778 Main St	Online	Cancelled	TRK69207593	5	
1197	275 Main St	Online	Delivered	TRK88039356	2	
1198	509 Main St	Debit Card	Cancelled	TRK71683331	4	
1199	201 Main St	Gift Card	Returned	TRK51116746	6	

	CouponCode	ReferralSource	TotalPrice
0	SAVE10	Instagram	2853.10
1	SAVE10	Referral	302.70
2	FREESHIP	Email	2753.40
3	SAVE10	Facebook	273.19
4	SAVE10	Email	2504.04
...	...	...	...
1195	FREESHIP	Google	107.04
1196	FREESHIP	Facebook	1325.06
1197	FREESHIP	Instagram	873.68
1198	WINTER15	Instagram	1050.08
1199	SAVE10	Referral	2242.32

[1200 rows x 14 columns]

```
In [12]: import pandas as pd

# Load dataset
df = pd.read_excel(r"C:\Users\Rinky Sharma\Downloads\Dataset for Data Analytics.xlsx")

# Fix Date column
# Try converting directly (handles strings like '2021-05-01' or numbers like 44930)
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')

# If your Date column is purely Excel serial numbers (like 44930), use:
# df['Date'] = pd.to_datetime(df['Date'], errors='coerce', unit='D', origin='1899-12-30')

# Ensure numeric columns
df['Quantity'] = pd.to_numeric(df['Quantity'], errors='coerce')
df['UnitPrice'] = pd.to_numeric(df['UnitPrice'], errors='coerce')
df['TotalPrice'] = pd.to_numeric(df['TotalPrice'], errors='coerce')

# Standardize text columns
df['Product'] = df['Product'].str.strip().str.title()
df['PaymentMethod'] = df['PaymentMethod'].str.strip().str.title()
df['OrderStatus'] = df['OrderStatus'].str.strip().str.title()
df['ReferralSource'] = df['ReferralSource'].str.strip().str.title()

# Save cleaned dataset
df.to_excel("Cleaned_Dataset.xlsx", index=False)

print(" Data cleaning complete. Cleaned dataset saved as Cleaned_Dataset.xlsx")
```

Data cleaning complete. Cleaned dataset saved as Cleaned_Dataset.xlsx